

Toothed whales

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Cetacea
SUBORDER	Odontoceti
FAMILIES	10
SPECIES	75

A much more diverse group than baleen whales, toothed whales make up almost 90 percent of all cetaceans. The group consists of dolphins (which include the killer whale), river dolphins, porpoises, white whales, sperm whales, and beaked whales. Most are medium-sized—although the sperm whale grows to up to 59 ft

(18 m)—and all possess teeth instead of baleen plates. On the forehead, there is a fluid-filled swelling called the melon, in front of which there is usually a beak. Most species are found around the world in deep water and coastal shallows, although a few live in freshwater. Some toothed whales migrate, but only the sperm whale travels long distances.

Anatomy

Most toothed whales have simple, conical, pointed teeth that are not divided into incisors, canines, premolars, and molars (as in most other mammals). Each tooth has a single root and is either straight or slightly curved. One set of teeth lasts the whale's lifetime. The number of teeth present varies from more than 40 pairs in each jaw in some dolphins, to a single pair (in the lower jaw) in beaked whales. Unlike

baleen whales, toothed whales have only one blowhole, which means that the skull is asymmetrical. The single blowhole usually opens at the top of the head (except in the sperm whale, where the opening is left of center at

the front of the skull). Another feature of many toothed whales is the streamlined head and the long, narrow beak seen in most species.

Feeding

While baleen whales trap their prey en masse, toothed whales capture victims individually. The even, conical teeth found in this group are perfect for catching slippery fish, which form the diet of most species. Sperm whales, however, mostly eat squid (but also other prey, such as octopus), and beaked whales have a mixed diet of fish and squid. Killer whales eat other whales (which they attack in packs), fish, and seals, which they sometimes seize from land, using waves to slide onto the beach. They also upset pack ice to knock prey into the water.

Fish-eating species (including killer whales) usually have numerous teeth to hold wriggling prey, while those that eat squid or octopus, for example,

have fewer teeth. Sperm whales have functional teeth only in the lower jaw (which is very narrow), and prey is held firmly between these teeth and the roughened palate in the upper jaw.

All toothed whales use echolocation (see p.244) to help find prey (and to avoid objects in their path).

Social behavior

Most toothed whales live in groups called pods or schools, which vary in size from less than 10 to over 1,000 (as seen in some dolphin species). The exact organization of these associations is poorly understood, although it is thought that subgroups form to perform independent tasks, such as feeding.

This suggests the presence of complex social structures. Some species, especially killer whales, appear to practice cooperative hunting, whereby prey is "herded" until trapped or surrounded. When swimming in formation, dolphins often leap out of the water, which may simply be playful behavior, or it may function as a form of communication.



STREAMLINED FOR SPEED

Atlantic spotted dolphins are fast, energetic swimmers, and, like several other toothed whale species, they often travel in large pods. The torpedo-shaped body is propelled through the water by powerful thrusts of the tail. The mottled coloration camouflages them from predators and prey in shallow, sunlit waters.